Importing and Executing a SysML Model

1.Importing from XMl

1. Select "SysML Connector" from the "Apps" tab in MATLAB.
2. Specify SystemModels/ElectricThrottleControlSysMLModel.xml as the source.
3. Specify the "Output Directory".
4. Click the "Import" button.
5. Verify the imported architecture model.

2. Model Execution and TestingModel

Execution:

1. Open SysMLArchModel/SysMLArchModel.prj.
2. Open ElectricThrottleControl.slx within the project.
3. Run the simulation and check the waveforms in the Data Inspector.

Test Execution:

1. Open TestFile/TestManagerFile.mldatx within the project.
2. Run the test cases in the Test Browser and verify the test results.

Try

If not yet installed, please install the "SysML Connector" from the following URL:
https://jp.mathworks.com/products/sysml.html

>>SystemModels/ElectricThrottleControlSysMLModel.xml
**　または、ElectricThrottleControlSysMLModel.mdzip**

>>SysMLArchModel/SysMLArchModel.prj

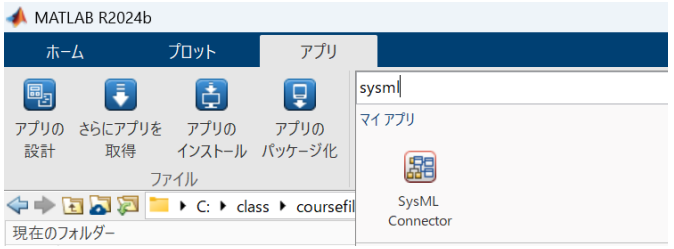
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Solution: Architecture and Model Creation (1)

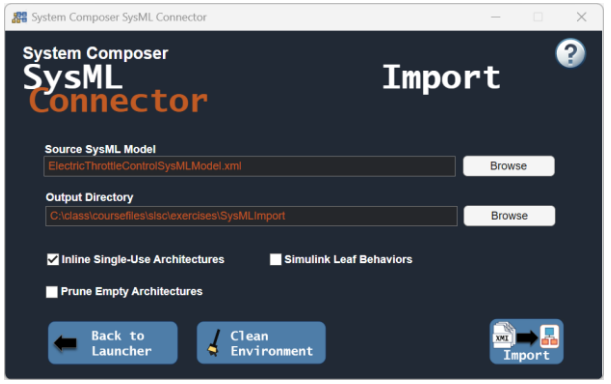
Try

>>ElectricThrottleControlSysMLModel.prj
>> ElectricThrottleControl.slx

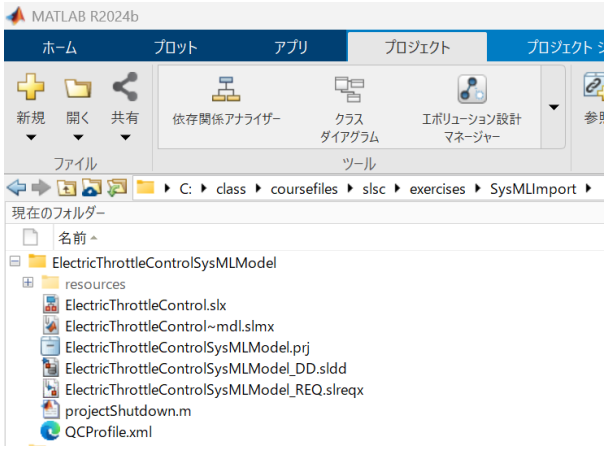
1.Importing from XMI
Select SysML Connector from the Apps tab.

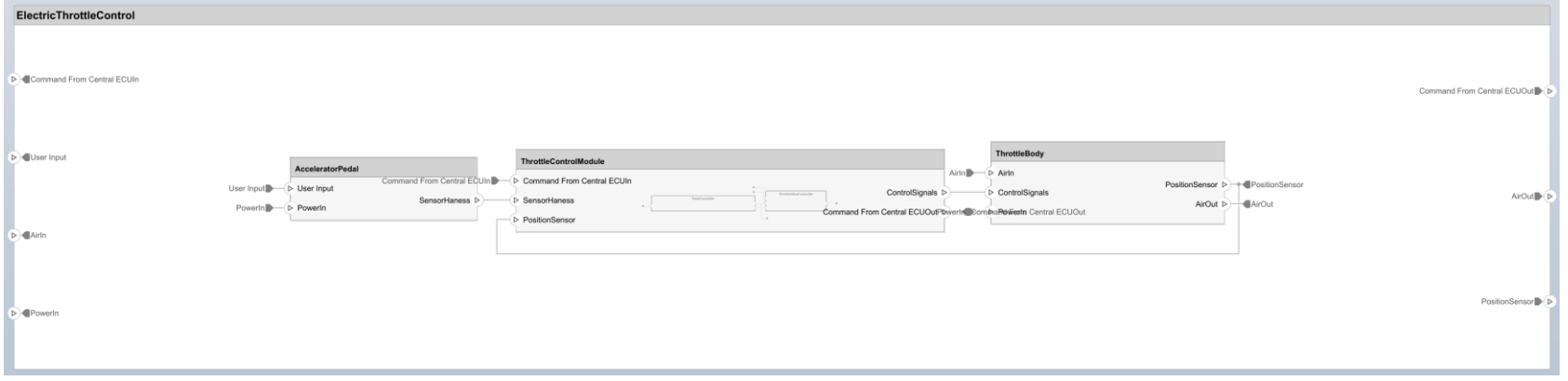


Import ElectricThrottleControlSysMLModel.xml.



Open the project file from the specified output directory and verify the imported architecture model.



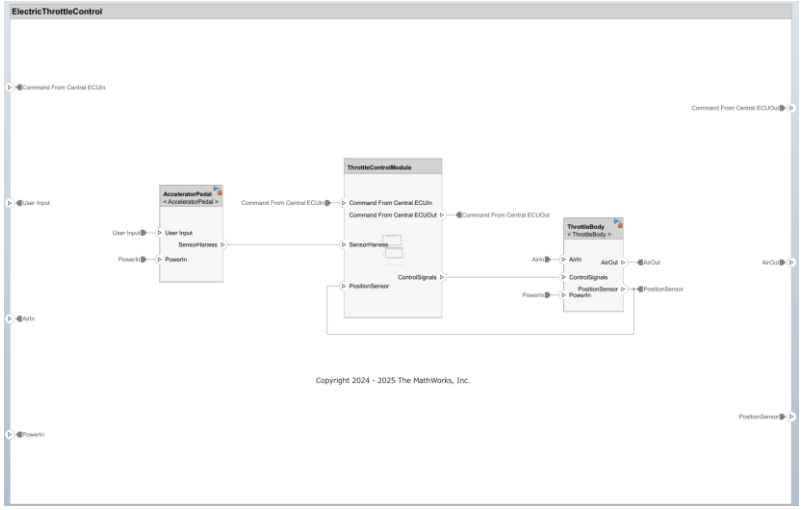


Solution: Architecture and Model Creation (2)

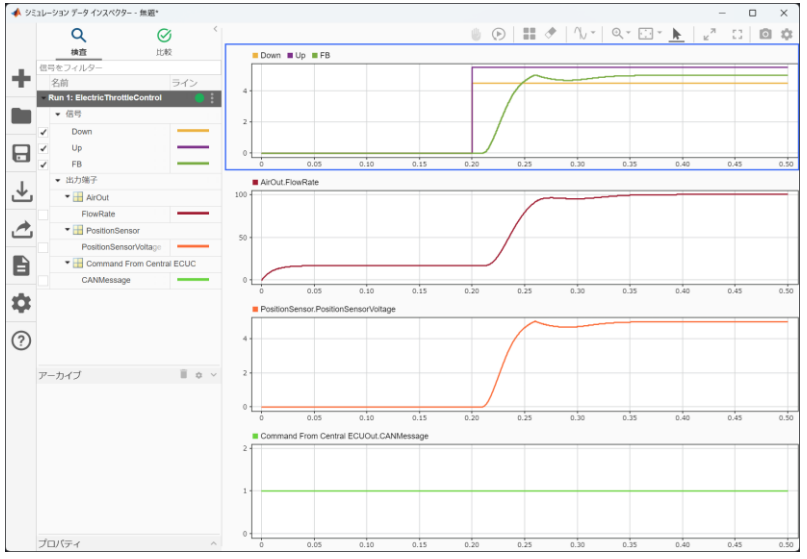
2. Model Execution and Testing

Model Execution:

Open ElectricThrottleControl.slx.



Run the simulation and check the waveforms in the Data Inspector.

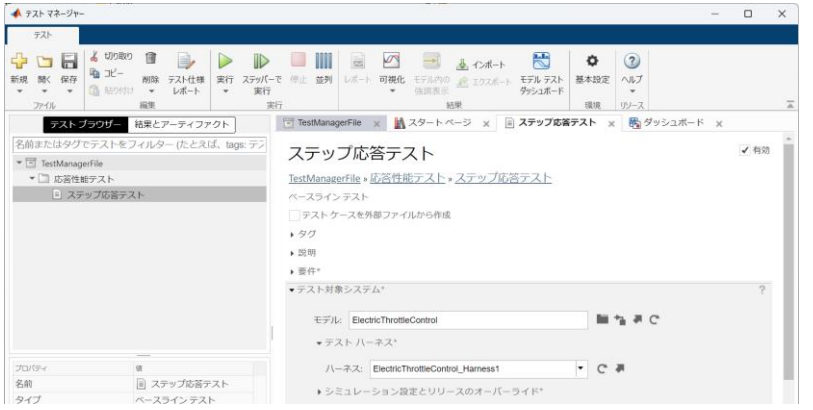


Try

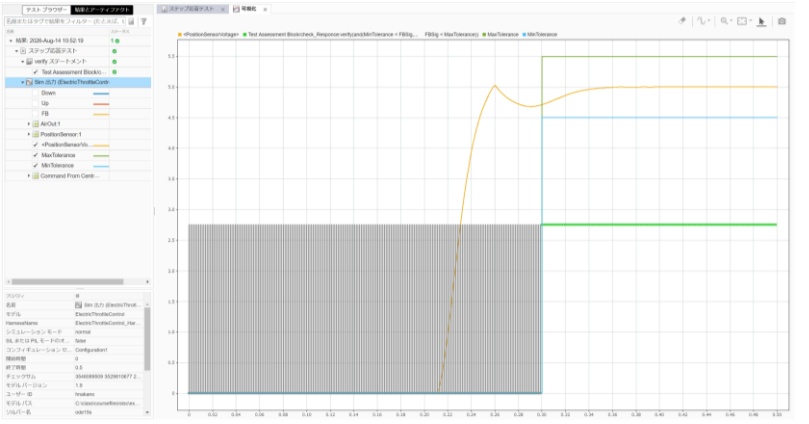
>>SysMLArchModel/SysMLArchModel.prj
>>TestFile/TestManagerFile.mldatx

Test Execution:

Run the test cases in the Test Browser.



Verify that the test requirements are met.



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Specifying Requirements through Detailed Design

While the requirements defined in the system design are met, the simulation confirms that a temporary drop occurs during the convergence process to the target value.Based on this result, response characteristics that were not defined by the high-level requirements alone are specified as detailed requirements.
Verification is performed based on the added requirements, and design parameters are reviewed while checking the simulation results.
Finally, re-verification is performed to ensure that both the existing and the newly added requirements are met.

1.Checking Simulation Results and Requirements

1.Open ElectricThrottleControl.slx within the project.
2.Run the simulation and check the PositionSensorVoltage waveform in the Data Inspector.Confirm that a non-monotonic response occurs during the convergence process to the target value.

2.Adding Detailed Requirements

1.Open the Requirements Editor and select "3 Response".
2.Add "3.1 Monotonic Response" under "3 Response".
3.Set the following requirement details:"After the response starts, the drop in PositionSensorVoltage from its previous maximum value must be 0.10V or less.“
4.Save the added requirements.

Try

>>SimDrivenReqRefinement/SysMLArchModel.prj

3. Testing Based on Added Requirements

1.Open TestFile/TestManagerFile.mldatx within the project.
2.Select "Monotonic Response Test" from the Test Browser.
3.Link the test with the "3.1 Monotonic Response" requirement.
4.Run the test case and confirm that the test fails against the added requirement.

4. Reviewing the Design via Simulation

1.Open ElectricThrottleControl.slx.
2.Change the PID controller parameters and run the simulation.
3.Compare the waveforms in the Data Inspector and select parameters that suppress the temporary drop while maintaining the existing response performance.

5. Re-testing Based on Requirements

1.Open TestFile/TestManagerFile.mldatx.
2.Run the test case again.
3.Confirm that the tests for both "3 Response" and "3.1 Monotonic Response" pass.
4.Review the simulation results and requirements to confirm that the insights gained from the detailed design have been reflected in the requirements and the design.

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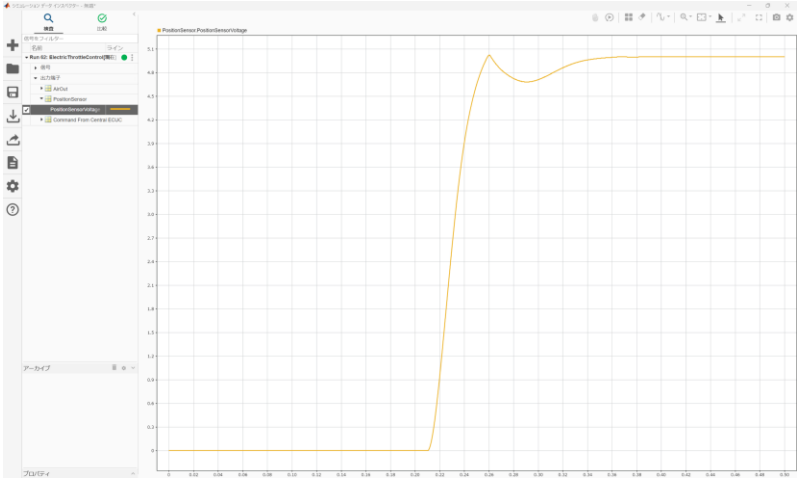
Solution: Specifying Requirements through Detailed Design (1)

Exercise

>>SimDrivenReqRefinement_ans

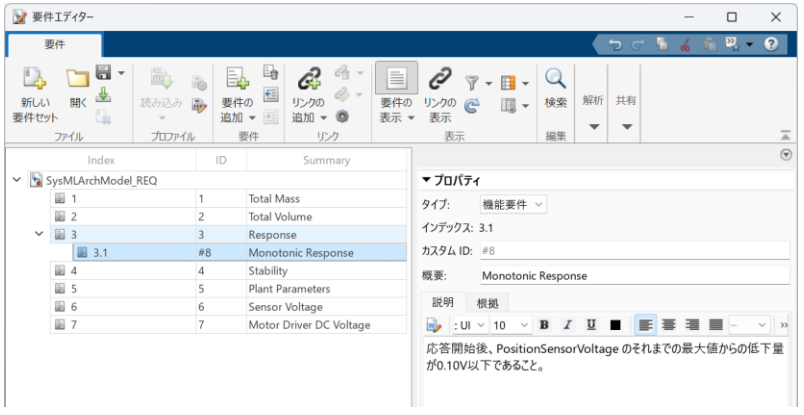
1.Checking Simulation Results and Requirements

Confirm that a temporary drop occurs during the convergence process to the target value.



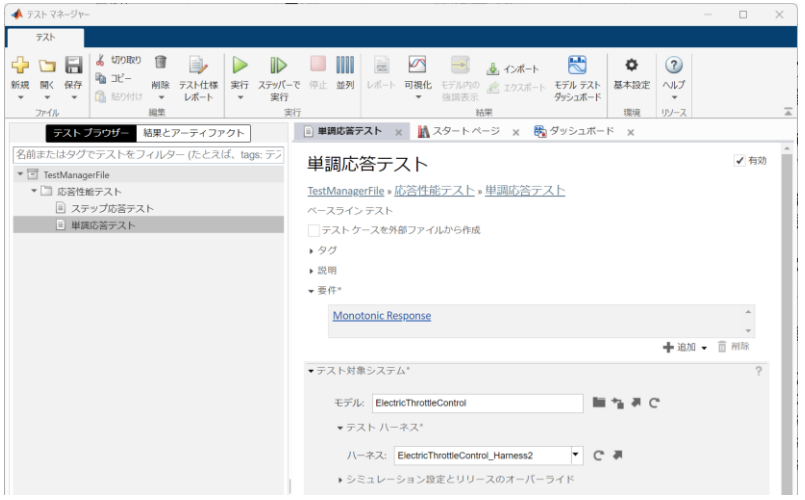
2. Adding Detailed Requirements

Set the following requirement details:

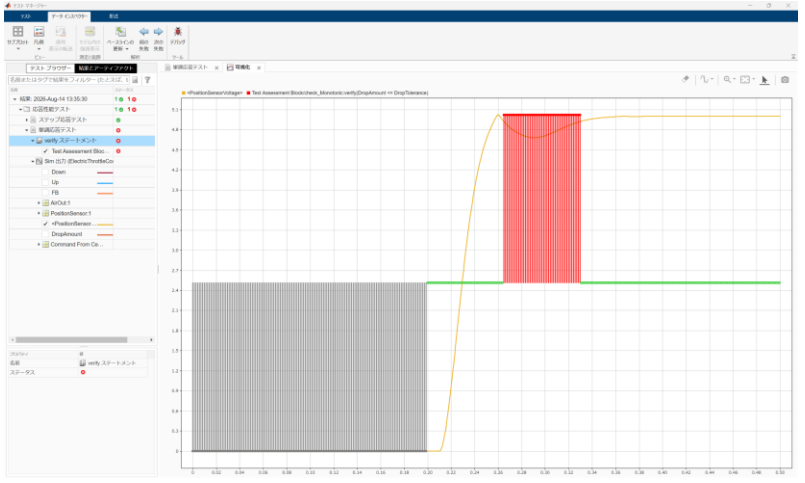


3. Testing Based on Added Requirements

Link the requirements and execute the test.



Confirm that the test fails against the added requirement.

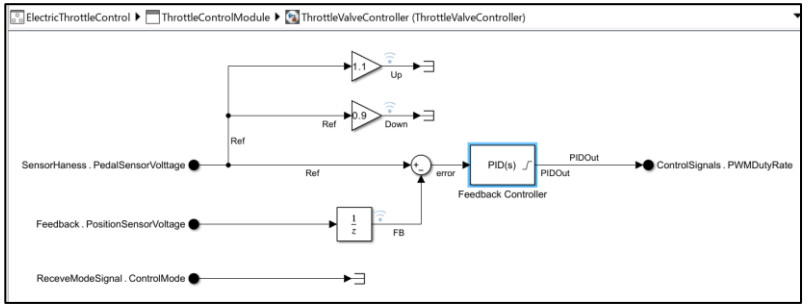


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Solution: Specifying Requirements through Detailed Design (2)

4. Reviewing the Design via Simulation

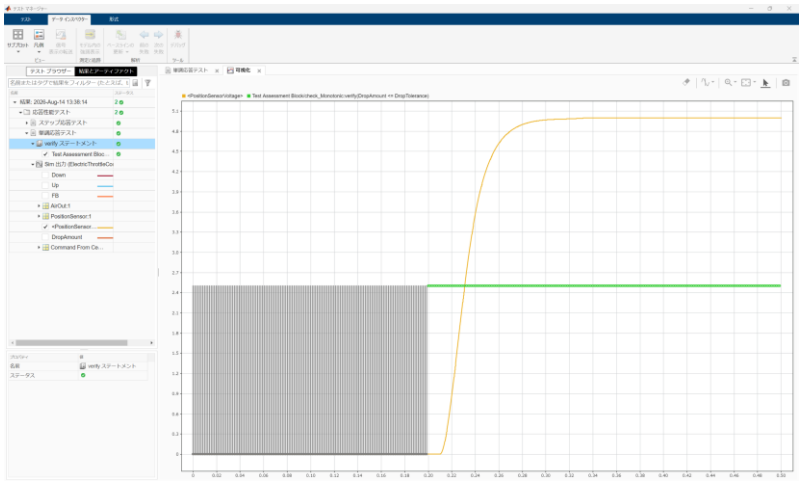
Change the parameters of the PID controller.





5. Re-testing Based on Requirements

Confirm that the test passes against the added requirement.



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